Daily Number Theory Drill #5

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Bounding Diophantine Equations · Inequality Logic · Factor Searches.

Section A Rapid Recognition ($\leq$15 s each)

Last digits, primes, and basic properties. Pure recognition.

1. Find the last digit of 3^{2025}.

Ans: 3

2. Factorise 1001 completely into primes.

Ans: $7 \times 11 \times 13$

3. Find $\gcd(42, 105)$.

Ans: 21

4. Find the number of positive divisors of 120.

Ans: 16

5. Evaluate $5^{10} \pmod 6$.

Ans: 1

6. Find the smallest positive integer x such that $2x \equiv 1 \pmod 5$.

Ans: 3

7. Find the smallest integer $n > 1$ such that $n \equiv 1 \pmod 3$ and $n \equiv 1 \pmod 4$.

Ans: 13

8. Write the prime factorisation of 360.

Ans: $2^3 \times 3^2 \times 5$

9. Calculate $7^3 \pmod{10}$.

Ans: 3

10. Find the largest prime factor of 9999.

Ans: 101

Section B Manipulation Drills ($\leq$50 s each)

Modular arithmetic, exponentiation, divisibility rules. Look for patterns.

1. The greatest power of 7 which is a factor of 50! is 7^k. What is k? *(after SMC 2023 Q12)*

Ans: 8

2. Find the smallest positive integer n such that $15n$ is a perfect cube.

Ans: 225

3. Find the remainder when $1! + 2! + 3! + \cdots + 100!$ is divided by 12.

Ans: 9

4. Which is larger, 3^{40} or 4^{30}?

Ans: 3^{40}

5. Find the smallest positive integer x such that $3x \equiv 4 \pmod{7}$.

Ans: 6

6. Find the last two digits of 7^{2022}.

Ans: 49

7. How many positive factors of 1000 are perfect squares?

Ans: 4

8. Find all integers x such that $0 \leq x < 8$ and $x^2 \equiv 1 \pmod{8}$.

Ans: 1, 3, 5, 7

9. Given $2^x 3^y = 12^3$, find $x + y$.

Ans: 9

10. Find all pairs of positive integers (x, y) such that $xy = x + y + 3$.

Ans: $(2, 5), (3, 3), (5, 2)$

Section C Substitution & Structure ($\leq$90 s each)

Synoptic links: use algebraic identities and combinatorial counting to solve these number theory problems.

1. The expression $\frac{n^2+3n+5}{n+1}$ takes integer values for certain positive integer values of n. What is the sum of all such positive integer values of n? *(after SMC 2023 Q19)*

Ans: 2

2. How many of the numbers 6, 7, 8, 9, 10 are factors of the sum $2^{2024} + 2^{2023} + 2^{2022}$? *(after SMC 2023 Q10)*

Ans: 2

3. Find all pairs of positive integers (x, y) such that $x^2 - y^2 = 17$.

Ans: (9, 8)

4. How many ordered pairs of positive integers (a, b) satisfy $\text{lcm}(a, b) = 300$?

Ans: 75

5. How many positive integers $n \leq 100$ have exactly 3 positive divisors?

Ans: 4

6. Find the sum of the digits of the largest integer n such that $n^3 + 100$ is a multiple of $n + 10$.

Ans: 17

7. Find all triples of positive integers (x, y, z) such that $x \leq y \leq z$ and $xyz = x + y + z$.

Ans: $(1, 2, 3)$

8. Find the sum of all integer values of n for which n^2+4n+3 is a perfect square. *(after AI Prompts D5)*

Ans: -4

Section D Challenge (TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Find all prime numbers p such that p^2+2 is also a prime number.

Ans: 3

2. Find all pairs of positive integers (x, y) such that $x^2y = x^2 + 3y + 2$.

Ans: $(2, 6)$

3. How many pairs of positive integers (x, y) with $x \leq y$ satisfy $\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$?

Ans: 5

4. Find the remainder when $1^{2025} + 2^{2025} + \cdots + 10^{2025}$ is divided by 11.

Ans: 0

5. Find the sum of all integers n for which $n^4 + 4$ is a prime number.

Ans: 0